

The language and notation of proof — Solutions

Warm-up

Solution to Question 1:

Given c : "I will eat ice cream" and d : "I will be happy".

(a) $d \Leftrightarrow c$ This is a biconditional statement.

I will be happy if and only if I eat ice cream.

(b) $c \Rightarrow d$ This is a conditional statement.

If I eat ice cream, then I will be happy.

(c) $c \vee d$ This is a disjunction.

I will eat ice cream or I will be happy.

Solution to Question 2:

The contrapositive of a statement "If P , then Q " is "If not Q , then not P ".

(a) Statement: "If I am happy, I will dance."

Contrapositive: If I do not dance, then I am not happy.

(b) Statement: "If my computer is powered, it will turn on."

Contrapositive: If my computer does not turn on, then it is not powered.

(c) Statement: "If the sun is outside, I will wake up."

Contrapositive: If I do not wake up, then the sun is not outside.

Solution to Question 3:

Translating mathematical symbols to English sentences:

(a) $\forall m \in \mathbb{R} \exists n \in \mathbb{R}$ such that $n = \cos(m)$

For every real number m , there exists a real number n such that n is the cosine of m .

(b) $\forall m \in \mathbb{Z}^+ \exists n \in \mathbb{Z}$ such that $m^3 = n$

For every positive integer m , there exists an integer n such that n is the cube of m .

(c) $\forall m \in \mathbb{Z}^+ \exists n \in \mathbb{Z}$ such that $n = -m$

For every positive integer m , there exists an integer n such that n is the negative of m .

Solution to Question 4:

To prove a statement by contrapositive, we prove "If not Q , then not P ".

(a) Statement: If a^2 is odd, then a is odd. Contrapositive: If a is even, then a^2 is even.

Let a be an even integer.

$$a = 2k \text{ for some } k \in \mathbb{Z}$$

$$a^2 = (2k)^2$$

$$a^2 = 4k^2$$

$$a^2 = 2(2k^2)$$

Since $2k^2 \in \mathbb{Z}$, a^2 is even.

(b) Statement: If $m - n$ is odd, then either m is odd or n is odd. Contrapositive: If m is even and n is even, then $m - n$ is even.

Let $m = 2k$ and $n = 2j$ for some $k, j \in \mathbb{Z}$.

$$m - n = 2k - 2j$$

$$m - n = 2(k - j)$$

Since $k - j \in \mathbb{Z}$, $m - n$ is even.

(c) Statement: If mn is even, then either m is even or n is even. Contrapositive: If m is odd and n is odd, then mn is odd.

Let $m = 2k + 1$ and $n = 2j + 1$ for some $k, j \in \mathbb{Z}$.

$$mn = (2k + 1)(2j + 1)$$

$$mn = 4kj + 2k + 2j + 1$$

$$mn = 2(2kj + k + j) + 1$$

Since $(2kj + k + j) \in \mathbb{Z}$, mn is odd.

Solution to Question 5:

Proof by contradiction assumes the negation of the statement and shows it leads to an impossibility.

(a) Statement: If it is raining, the ground is wet. Assumption: It is raining, but the ground is not wet.

If it is raining, water falls from the sky onto the ground.

If the ground is not wet, no water has touched it.

This is a contradiction because water cannot simultaneously fall on the ground and not touch it.

(b) Statement: If you turn off the lights, the room is dark. Assumption: You turn off the lights, but the room is not dark.

If the room is not dark, there must be a light source present.

However, you turned off the lights and there are no other light sources.

This is a contradiction because there cannot be light in a room with no light sources.

(c) Statement: If you drop ice into a warm drink, it gets colder. Assumption: You drop ice into a warm drink, but the drink does not get colder.

Ice absorbs heat from the warmer liquid to melt.

If the drink does not get colder, it means no heat was removed from the liquid.

This is a contradiction because heat naturally flows from hot to cold.

Skill-building

Solution to Question 1:

Given c : "I will run today" and d : "I will be fast".

(a) $d \wedge \neg c$

I will be fast and I will not run today.

(b) $\neg d \Rightarrow c$

If I am not fast, then I will run today.

(c) $c \not\Rightarrow \neg d$

Running today does not imply that I will not be fast.

Solution to Question 2:

The contrapositive of "If P , then Q " is "If not Q , then not P ".

(a) Statement: "If I am happy, I will dance."

Contrapositive: If I do not dance, then I am not happy.

(b) Statement: "If my computer is powered, it will turn on."

Contrapositive: If my computer does not turn on, then it is not powered.

(c) Statement: "If the sun is outside, I will wake up."

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Question 3:

Translating mathematical symbols to English sentences:

(a) $\forall m \in \mathbb{R} \exists n \in \mathbb{R}$ such that $n = \cos(m)$

For every real number m , there exists a real number n such that n is the cosine of m .

(b) $\forall m \in \mathbb{Z}^+ \exists n \in \mathbb{Z}$ such that $m^3 = n$

For every positive integer m , there exists an integer n such that n is the cube of m .

(c) $\forall m \in \mathbb{Z}^+ \exists n \in \mathbb{Z}$ such that $n = -m$

For every positive integer m , there exists an integer n such that n is the negative of m .

Solution to Question 4:

To prove a statement by contrapositive, we prove "If not Q , then not P ".

(a) Statement: If $a^2 + 2a + 6$ is even, then a is even. Contrapositive: If a is odd, then $a^2 + 2a + 6$ is odd.

Let $a = 2k + 1$ for some $k \in \mathbb{Z}$.

$$a^2 + 2a + 6 = (2k + 1)^2 + 2(2k + 1) + 6$$

$$a^2 + 2a + 6 = (4k^2 + 4k + 1) + (4k + 2) + 6$$

$$a^2 + 2a + 6 = 4k^2 + 8k + 9$$

$$a^2 + 2a + 6 = 2(2k^2 + 4k + 4) + 1$$

Since $(2k^2 + 4k + 4) \in \mathbb{Z}$, $a^2 + 2a + 6$ is odd.

(b) Statement: If $m - n$ is odd, then either m is odd or n is odd. Contrapositive: If m is even and n is even, then $m - n$ is even.

Let $m = 2k$ and $n = 2j$ for some $k, j \in \mathbb{Z}$.

$$m - n = 2k - 2j$$

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Since $k - j \in \mathbb{Z}$, $m - n$ is even.

(c) Statement: If mn is even, then either m is even or n is even. Contrapositive: If m is odd and n is odd, then mn is odd.

Let $m = 2k + 1$ and $n = 2j + 1$ for some $k, j \in \mathbb{Z}$.

$$mn = (2k + 1)(2j + 1)$$

$$mn = 4kj + 2k + 2j + 1$$

$$mn = 2(2kj + k + j) + 1$$

Since $(2kj + k + j) \in \mathbb{Z}$, mn is odd.

Solution to Question 5:

Proof by contradiction assumes the negation of the statement and shows it leads to an impossibility.

(a) Statement: $\sqrt{2}$ is irrational. Assumption: $\sqrt{2}$ is rational.

Let $\sqrt{2} = \frac{p}{q}$ where $p, q \in \mathbb{Z}$, $q \neq 0$, and $\gcd(p, q) = 1$.

$$2 = \frac{p^2}{q^2}$$

$$p^2 = 2q^2$$

Since p^2 is even, p must be even. Let $p = 2k$.

$$(2k)^2 = 2q^2$$

$$4k^2 = 2q^2$$

$$q^2 = 2k^2$$

Since q^2 is even, q must be even.

This contradicts $\gcd(p, q) = 1$.

(b) Statement: The sum of a rational and irrational number is irrational. Assumption: The sum of a rational r and an irrational i is a rational s .

Let $r = \frac{a}{b}$ and $s = \frac{c}{d}$ where $a, b, c, d \in \mathbb{Z}, b, d \neq 0$.

$$r + i = s$$

$$i = s - r$$

$$i = \frac{c}{d} - \frac{a}{b}$$

$$i = \frac{bc - ad}{bd}$$

Since $bc - ad$ and bd are integers, i is rational.

This contradicts the assumption that i is irrational.

(c) Statement: $\log_3 5$ is irrational. Assumption: $\log_3 5$ is rational.

Let $\log_3 5 = \frac{p}{q}$ where $p, q \in \mathbb{Z}^+, q \neq 0, \gcd(p, q) = 1$.

$$3^{p/q} = 5$$

$$(3^{p/q})^q = 5^q$$

$$3^p = 5^q$$

Since 3^p is always odd and 5^q is always odd, this doesn't immediately show a contradiction via parity. However, by the Fundamental Theorem of Arithmetic, a power of 3 cannot equal a power of 5. This is a contradiction.

Solution to Question 6:

Evaluating the truth value of statements:

(a) For any 2 numbers s and t , $|s + t| \leq |s| + |t|$.

This is the Triangle Inequality. It is TRUE.

(b) For any number x , $x^2 > x$.

This is FALSE.

Counterexample: Let $x = 0.5$.

$$(0.5)^2 = 0.25.$$

$$0.25 \not> 0.5.$$

(c) For any 2 numbers a and b , If $a < b$ then $|a| < |b|$.

This is FALSE.

Counterexample: Let $a = -5, b = 2$.

$-5 < 2$ is true, but $|-5| < |2|$ is $5 < 2$, which is false.

Easier Exam Questions

Solution to Question 1:

Slogan: "If you are a strong swimmer, then you are unlikely to drown." Let P : "you are a strong swimmer" and Q : "you are unlikely to drown".

The statement is $P \Rightarrow Q$.

The logically equivalent statement is the contrapositive: $\neg Q \Rightarrow \neg P$.

$\neg Q$: "you are likely to drown"

$\neg P$: "you are not a strong swimmer"

Contrapositive: "If you are likely to drown, then you are not a strong swimmer."

Correct Option: D

Solution to Question 2:

Statement: "If x is even, then $5x - 11$ is odd." Let P : " x is even" and Q : " $5x - 11$ is odd".

The contrapositive is $\neg Q \Rightarrow \neg P$.

$\neg Q$: " $5x - 11$ is even"

$\neg P$: " x is odd"

Contrapositive: "If $5x - 11$ is even, then x is odd."

Correct Option: B

Solution to Question 3:

Statement: "I listen to music and I play games." Let P : "I listen to music" and Q : "I play games".

The statement is $P \wedge Q$.

The negation is $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$ (by De Morgan's Law).

$\neg P$: "I don't listen to music"

$\neg Q$: "I don't play games"

Negation: "I don't listen to music or I don't play games."

Correct Option: A

Solution to Question 4:

Statement: "If n is a positive integer, $3^n - 1$ is divisible by 4". This is a conditional statement $P \Rightarrow Q$.

The negation of $P \Rightarrow Q$ is $P \wedge \neg Q$.

P : " n is a positive integer"

$\neg Q$: " $3^n - 1$ is NOT divisible by 4"

Negation: " n is a positive integer BUT $3^n - 1$ is NOT divisible by 4."

Correct Option: C

Solution to Question 5:

Evaluating the truth of the statements:

(A) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $xy = 10$

If $x = 0$, there is no y such that $0 \cdot y = 10$. Therefore, this is FALSE.

(B) $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $y = x^2$

For any real number x , its square x^2 is always a real number. Let $y = x^2$. Therefore, this is TRUE.

(C) $\exists x \in \mathbb{R}$, such that $\forall y \in \mathbb{R}, xy = 10$

This would mean one specific x works for all y .

If $y = 1, x = 10$. If $y = 2, x = 5$. Since x must be constant, this is FALSE.

(D) $\exists x \in \mathbb{R}$, such that $\forall y \in \mathbb{R}, y = x^2$

This would mean all y are equal to the same square x^2 . This is FALSE.

Correct Option: B

Harder Exam Questions

Solution to Question 6:

Given $x, y \in \mathbb{Z}, x, y \geq 0$.

(A) $\exists x(\forall y : y = 1 - 3x)$

This says there is one x that makes $y = 1 - 3x$ for all y .

This is impossible because y is a variable. Therefore, this is FALSE.

(B) $\exists x(\exists y : y = 5 + 2x)$

Let $x = 0$, then $y = 5$. Since $0, 5 \in \mathbb{Z}^+$, this is TRUE.

(C) $\forall x(\exists y : y = x^2)$
the

For any $x \in \mathbb{Z}^+$, its square x^2 is also in $\mathbb{Z}^+$. Therefore, this is TRUE.

(D) $\forall x(\exists y : y - x = 0)$

For any x , we can choose $y = x$. Since $x \in \mathbb{Z}^+$, then $y \in \mathbb{Z}^+$. Therefore, this is TRUE.

Correct Option: A

Solution to Question 7:

Statement: "For all positive integers n , there exists a positive integer p such that $p = n^2$ "

For all positive integers $n \implies \forall n \in \mathbb{Z}^+$

There exists a positive integer $p \implies \exists p \in \mathbb{Z}^+$

such that $p = n^2$

Combining these: $\forall n \in \mathbb{Z}^+ \exists p \in \mathbb{Z}^+$ such that $p = n^2$

Correct Option: C

Solution to Question 8:

Let the three-digit number be represented as $100x + 10y + z$. Given $x + z = y$.

$$\begin{aligned}
\text{The number } N &= 100x + 10y + z \\
N &= 100x + 10(x + z) + z \\
N &= 100x + 10x + 10z + z \\
N &= 110x + 11z \\
N &= 11(10x + z)
\end{aligned}$$

Since $10x + z$ is an integer, N is divisible by 11.

Solution to Question 9:

We want to find positive integers a, b, c such that $a^2 + b^2 + c^2 = 14$. Assume without loss of generality that $a \leq b \leq c$.

Possible squares less than 14 are: 1, 4, 9.

If $c^2 = 9$, then $a^2 + b^2 = 14 - 9 = 5$.

The only squares that sum to 5 are 1 and 4.

Thus, $a^2 = 1, b^2 = 4, c^2 = 9$.

This gives $a = 1, b = 2, c = 3$.

If $c^2 = 4$, then $a^2 + b^2 = 14 - 4 = 10$.

The only squares that sum to 10 are 1 and 9, but we assumed c was the largest, so c^2 must be ≥ 9 .

If $c^2 = 1$, then $a^2 + b^2 = 13$, which is impossible for $a, b \leq 1$.

Thus, the only positive integers are 1, 2, and 3.

Solution to Question 10:

Proof by contradiction: Assume there are a finite number of prime numbers.

Let the set of all primes be $P = \{p_1, p_2, \dots, p_n\}$.

Construct a number $Q = (p_1 \cdot p_2 \cdot \dots \cdot p_n) + 1$.

By the Fundamental Theorem of Arithmetic, Q must have at least one prime factor q .

If q is one of the primes in our set P , then q must divide Q .

However, Q divided by any p_i leaves a remainder of 1.

Therefore, q cannot be any of the primes in P .

This means q is a prime not in our finite set P .

This is a contradiction to the assumption that P contains all primes.

Therefore, there are an infinite number of prime numbers.